

2 218 514 FinalIII-Phase2- Report (1) output (1) (1).docx

by turnitin student

Submission date: 11-Apr-2025 09:51AM (UTC-0400)

Submission ID: 2642495826

File name: 2_218_514_FinalIII-Phase2-Report_1_output_1_1_.docx (1.47M)

Word count: 3974

Character count: 23264

DETECTING CELLS RESPONSIBLE FOR SKIN CANCER DEVELOPMENT

PHASE II REPORT

Submitted by

SABARIMANI S

(2116210701218)

KAJA MOIDHEEN P

(2116210701514)

¹
in partial fulfillment for the award of the

degree of

BACHELOR OF ENGINEERING

in

COMPUTER SCIENCE AND ENGINEERING



RAJALAKSHMI ENGINEERING

COLLEGE ANNA UNIVERSITY,

CHENNAI

DECEMBER 2024

RAJALAKSHMI ENGINEERING COLLEGE, CHENNAI

BONAFIDE CERTIFICATE

Certified that this Project titled **“DETECTING CELLS RESPONSIBLE FOR SKIN CANCER DEVELOPMENT”** is the bonafide work of **“SABARIMANI S (2116210701218), KAJAMOIDEEN (2116210701514)”** ¹ who carried out the work under my supervision. Certified further that to the best of my knowledge the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on this or any other candidate.

SIGNATURE

Dr.P. Kumar.,

Director, Computing and Information
Science & HOD
Department of Computer Science
and Engineering,
Rajalakshmi Engineering College,
Chennai - 602 105.

SIGNATURE

Dr.S. Anantha Sivaprakasam

Professor
Department of Computer Science
and Engineering,
Rajalakshmi Engineering
College,
Chennai - 602 105.

Submitted to Project Viva-Voce Examination held on _____

Internal Examiner

External Examiner

ABSTRACT

Skin cancer is one of the most prevalent cancers worldwide, making early detection crucial for effective treatment. This work proposes a deep learning-based method for skin cancer classification using the VGG19 convolutional neural network (CNN), building upon previous research involving ResNet-50. The system is developed using the HAM10000 dataset, which contains over 9,000 labeled images. To address class imbalance, preprocessing techniques such as data augmentation and the Synthetic Minority Oversampling Technique (SMOTE) are applied. The VGG19 model, pre-trained on ImageNet, is fine-tuned to classify seven types of skin lesions with enhanced accuracy. Additionally, a Python-based graphical user interface (GUI) using Tkinter allows users to upload skin lesion images for real-time classification. Experimental results indicate that VGG19 outperforms ResNet-50 in terms of accuracy, overfitting reduction, and generalization. The proposed system serves as an automated, accessible, and non-invasive diagnostic tool, assisting both medical professionals and individuals in the early detection of skin cancer. By improving lesion classification accuracy, this approach contributes to more reliable skin cancer detection, facilitating early intervention and ultimately improving patient outcomes.

ACKNOWLEDGMENT

First, we thank the almighty god for the successful completion of the project. Our sincere thanks to our chairman **Mr. S. Meganathan** for his sincere endeavor in educating us in his premier institution. We would like to express our deep gratitude to our beloved Chairperson **Dr. Thangam Meganathan** for her enthusiastic motivation which inspired us a lot in completing this project and Vice Chairman **Mr. Abhay Shankar Meganathan** for providing us with the requisite infrastructure. We also express our sincere gratitude to our college Principal, **Dr. S. N. Murugesan** and **Dr. P. Kumar Director, Computing and Information Science and HOD, Department of Computer Science and Engineering** and our project mentor **Dr. Anantha Sivaprakasam, Professor** for his encouragement and guiding us throughout the project. We are very glad to thank our Project Coordinator, **Dr. Rakesh Kumar M, Assistant Professor** for his useful tips during our review to build our project. We would like to thank our parents, friends, all faculty members and supporting staff for their direct and indirect involvement in successful completion of the project for their encouragement and support.

SABARIMANI S 2116210701218

KAJA MOIDHEEN 2116210701514

1
TABLE OF CONTENTS

CHAPTER NO.	TITLE	PAGE NO.
	ABSTRACT	iii
	LIST OF TABLES	vii
	LIST OF FIGURES	viii
	LIST OF ABBREVIATIONS	ix
1.	INTRODUCTION	1
	1.1 GENERAL	1
	1.2 OBJECTIVES	2
	1.3 EXISTING SYSTEM	2
2.	LITERATURE SURVEY	3
3.	PROPOSED SYSTEM	14
	3.1 GENERAL	14
	3.2 SYSTEM ARCHITECTURE DIAGRAM	14
	3.3 DEVELOPMENT ENVIRONMENT	15
	3.3.1 HARDWARE REQUIREMENTS	15
	3.3.2 SOFTWARE REQUIREMENTS	16
	3.4 DESIGN THE ENTIRE SYSTEM	16
	3.2 CLASS DIAGRAM	17
	3.3 DATA FLOW DIAGRAM	18
	3.4 ACTIVITY DIAGRAM	19
4.	MODULE DESCRIPTION	20

	4.1 SYSTEM ARCHITECTURE	20
	4.1.1 USER INTERFACE DESIGN	20
	4.1.2 BACK END INFRASTRUCTURE	21
	4.2 DATA COLLECTION & PREPROCESSING	21
	4.2.1 DATASET & DATA LABELLING	21
	4.2.2 DATA PREPROCESSING	21
	4.2.3 FEATURE SELECTION	22
	4.2.4 CLASSIFICATION & MODEL SELECTION	22
	4.2.5 PERFORMANCE EVALUATION	23
	4.2.6 MODEL DEPLOYMENT	23
5.	IMPLEMENTATIONS AND RESULTS	24
	5.1 IMPLEMENTATION	24
	5.2 OUTPUT SCREENSHOTS	24
6.	CONCLUSION AND FUTURE ENHANCEMENT	
	6.1 CONCLUSION	29
	6.2 FUTURE ENHANCEMENT	29
	REFERENCES	31

LIST OF TABLES

TABLE NO	TITLE	PAGE NO
3.1	HARDWARE REQUIREMENTS	13
3.2	SOFTWARE REQUIREMENTS	14
3.3	COMPARISON OF FEATURES	19

LIST OF FIGURES

FIGURE NO	TITLE	PAGE NO
3.1	SYSTEM ARCHITECTURE	15
3.2	ACTIVITY DIAGRAM	17
3.3	DFD DIAGRAM	18
5.1	DATA PREPROCESSING AND FEATURE SELECTION	26
5.2	PERFORMANCE EVALUATION & OPTIMIZATION	27

⁹
LIST OF ABBREVIATIONS

S.No	ABBR	Expansion
1	AI	Artificial Intelligence
2	CSS	Cascading Style Sheet
3	DFD	Data Flow Diagram
4	RESNET	RESIDUAL NETWORK
5	ML	Machine Learning
6	VGG	Visual Geometry Group

14 CHAPTER 1

INTRODUCTION

1.1 GENERAL

15
Skin cancer-the most common cancer in the world-is characterized by uncontrolled growth of abnormal skin cells. Although it highly correlates to the period that elapses over an extended exposure to ultraviolet (UV) radiation, it is possible to occur anywhere in the body not exposed to the sun. There are three main types of cancers that vary in severity and treatment requirements: 7 basal cell carcinoma, squamous cell carcinoma, and melanoma. Early detection is crucial, but the conventional methods in diagnosing encompass dermatological visual inspections and, more often than not, biopsies that are subjective and time consuming and require visits to dermatologists; a commodity sadly lacking in many remote or underdeveloped regions. New breakthroughs in artificial intelligence, particularly deep learning, open up new avenues for improving medical diagnostics. In the area of image analysis, CNNs have done pretty well. As a result, automatic skin cancer detection becomes one of the typical applications for this approach, and hence, this paper focuses on using a deep CNN model, VGG19, to detect skin lesions. The system uses a large labelled set of images for diagnosis, which reduces the necessity of hand-checks and improves on current approaches. The project aims to identify fast, accurate non-invasive diagnostics. It would make early detection easier for many people-a factor that improves outcomes for the patient and helps practices with cancer treatment.

1.2 OBJECTIVE

The primary reason for implementing deep learning technology in skin cancer identification is to produce accurate and efficient diagnostic tools. Automation can be cost effective and saves time compared with doctors doing it manually. Deep learning models promote effective and rapid detection of skin cancer thereby assisting in early intervention and better patient results while supporting the healthcare professionals in their decision-making process.

1.3 EXISTING SYSTEM

In Traditional methods of skin cancer detection rely heavily on the visual inspection of dermatologists and biopsies, which may cause inconsistencies and delays in diagnosis. To overcome this, CAD systems have been developed, using handcrafted features for classification. However, these systems require a lot of human intervention and lack the flexibility needed for more accurate and adaptive detection. In previous work, a deep learning-based approach with ResNet-50 was utilized, achieving a high accuracy of classification. However, although it was successful, ResNet-50 still encounters problems when dealing with more complex features in skin lesions, as it experiences vanishing gradients and is less effective in deeper feature extraction.

This calls for even more sophisticated CNN architectures that will improve the accuracy and reliability of skin cancer detection. This has led to the exploration of models like VGG19, which offers a deeper and more structured architecture for improved feature extraction. VGG19's design allows for more effective handling of intricate patterns and details in skin lesions, addressing the limitations of ResNet-50. By using VGG19, it is possible to improve classification performance and create a more robust and adaptable system for skin cancer detection. This research aims to leverage VGG19's capabilities to enhance diagnostic accuracy and provide a more reliable tool for early skin cancer detection, reducing the potential for misdiagnosis and improving patient outcomes.

CHAPTER 2

LITERATURE SURVEY

In this paper [1] mapping strategies for the control of skin cancer will be discussed at the clinico-genetic-environmental interface. The principles presented are centered around personalized treatments that embody both conventional and alternative therapies and emphasize a multi-player team approach for maximal patient outcomes.

Mitigation of UV radiation [2] exposure is one of the study's strategies into risk factors causing skin cancers. This will include different methodologies of measuring UV exposure and public health campaigns to reduce the occurrence of skin cancers.

In this review [3] of the use of Convolutional Neural Networks (CNNs) in the detection of skin cancer, especially melanoma. The paper discusses different CNN designs used in image classification.

This review paper [4] deals with the early diagnosis of melanomas by dermoscopic imaging, histopathological analysis, and automated systems. According to the paper, early detection is the strong predictor of survival, and then further study reflects on challenges facing such technologies in clinical practices.

This new research [5] evaluates the efficacy of preventive measures, such as educations and screenings programs, public policies aimed at reducing skin cancer incidence, and whether these programs were positive for other populations.

This article [6] discusses the application of techniques of image processing combined with machine learning models for melanoma detection. Techniques of preprocessing such as segmentation followed by feature extraction in the diagnosis of melanoma.

This paper [7] Focuses on recent advances in melanoma skin cancer image processing. The paper reviews methods such as color normalization, edge detection, and feature extraction correlated to classification algorithms in order to work towards the improvement of efficiency and accuracy of melanoma detection.

This paper [8] highlights that the texture features of the images have a strong role in dermoscopic melanoma detection. In this case, first-order statistical methods, like mean, variance, and skewness, are utilized to select the principal features which aid in discriminating melanoma from benign lesions.

The paper [9] presents an expert system based on machine vision for the automated detection of skin cancer. It focuses on the system's ability to process images of skin lesions, analyze them for potential cancerous features, and provide diagnostic recommendations without human intervention.

In this paper [10] is proposing a hybrid skin cancer detection approach by combining CNN with texture features obtained from dermoscopy images. Combining these techniques, the application is really able to enhance the classification of benign and malignant skin lesions more accurately.

In this paper[11] discusses how color enhancement techniques impact the earlier diagnosis of skin cancer using CNNs. It is reported that by enhancing the color contrast in dermoscopic images, the presence of skin cancer.

This investigates [12] of the application of deep learning algorithms towards detection of skin cancer. It puts forth the fact that how CNNs classify as data imbalance.

This paper [13] is about the classifying of dermoscopic images for early detection of skin cancer. The paper elaborately reviews several machine learning classifiers, including SVM and CNNs, that are used to identify dermoscopic images and the challenge in increasing their accuracy.

This paper [14] introduces a hybrid technique that integrates feature fusion with CNNs for melanoma detection. Since the model had already integrated some features obtained from different domains like color, texture, and shape, its resulting classification performance is good for early-stage detection of melanoma.

The paper [15] focuses on the non-invasive imaging of skin cancer detection, and its concentration lies in dermoscopy with OCT. This paper is discussing accuracy and practicality of these techniques in comparison to invasive biopsies for melanoma diagnosis.

This study [16] on conventional verses CNN-based approaches for skin cancer diagnosis, efficiency, cost, and accuracy. The feasibility of such systems in low-resource settings and what must be sacrificed in terms of system complexity for the detection accuracy is also depicted in the paper.

In this article [17] suggests the use of a new hybrid artificial neural network (ANN)

model that involves color space transformations such as RGB to HSV that can improve the diagnostic accuracy of skin cancer.

In this paper [18] presents a multi-model learning system combining several deep learning models to learn effectively for the detection of skin cancer. The system can then precisely determine as to whether it is melanoma or non-melanoma skin cancer from the various types and sources of data in the form of clinical images and patient history.

This paper [19] discusses the hardware implementation of the OTSU thresholding method which is one of the most popular image segmentation techniques for skin cancer detection. It assesses the computational efficiency and accuracy of OTSU thresholding in real-time processing systems with dermoscopy images.

This paper [20] explores the use of the ABCD rule (Asymmetry, Border, Color, and Diameter) for the analysis of dermoscopy images to detect early signs of skin cancer. It combines traditional dermatological criteria with modern image processing techniques to improve diagnostic accuracy.

PROPOSED SYSTEM

3.1 GENERAL

The proposed system uses VGG19, a 19-layer ¹⁶ deep Convolutional Neural Network (CNN), ¹⁰ to enhance the accuracy of skin cancer classification. Unlike ResNet-50, which relies on residual connections, VGG19 uses smaller convolutional filters (3×3) that allow it to capture finer details in ² skin lesions. The model is trained and validated using the HAM10000 dataset, which contains a diverse set of skin lesion images. With regards to class imbalance, the above techniques of synthetic minority oversampling technique (SMOTE) along with data augmentation are used. The system exploits transfer learning through the fine-tuning of a pre-trained VGG19 model on the HAM10000 dataset. This is helpful in saving training time with high accuracy. Fine-tuning enables the model to adapt to the particular skin lesion characteristics, thus improving the ability of the model to classify different types of skin cancer. A graphical user interface (GUI) developed with Tkinter gives a friendly platform for real-time classification, hence making the system accessible to medical professionals and the general public. This system is intended to be more accurate, efficient, and accessible for skin cancer detection by incorporating VGG19 and advanced preprocessing techniques. Real-time classification in a GUI ensures that the tool is practical for everyday use, supporting early detection and improving overall diagnostic outcomes.

3.2 SYSTEM ARCHITECTURE DIAGRAM

The system architecture Fig 3.1 begins by pre-processing the dataset which includes data augmentation and rescaling into arrays. To handle class imbalances, SMOTE is applied to the dataset that enhances the same. For this preprocessed data, the VGG19 is trained and tested, which gives it the best performance. A GUI built with Tkinter in Python will enable uploading images of lesions, and the system is capable of providing predictions for the type of skin cancer within a split of seconds.

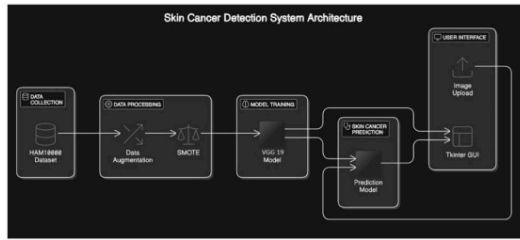


Fig 3.1: System Architecture

3.3 DEVELOPMENTAL ENVIRONMENT

3.3.1 HARDWARE REQUIREMENTS

The hardware specifications could be used as a basis for a contract for the implementation of the system. This therefore should be a full, full description of the whole system. It is mostly used as a basis for system design by the software engineers.

Table 3.1 Hardware Requirements

COMPONENTS	SPECIFICATION
PROCESSOR	Intel Core i3
RAM	4 GB RAM
POWER SUPPLY	+5V power supply

3.3.2 SOFTWARE REQUIREMENTS

The software requirements paper contains the system specs. This is a list of things which the system should do, in contrast from the way in which it should do things. The software requirements are used to base the requirements. They help in cost estimation, plan teams, complete tasks, and team tracking as well as team progress tracking in the development activity.

¹
Table 3.2 Software Requirements

COMPONENTS	SPECIFICATION
Operating System	Windows 7 or higher
Frontend	HTML,CSS
Backend	Python
Dataset	HAM10000

3.4 DESIGN OF THE ENTIRE SYSTEM

3.4.1 CLASS DIAGRAM

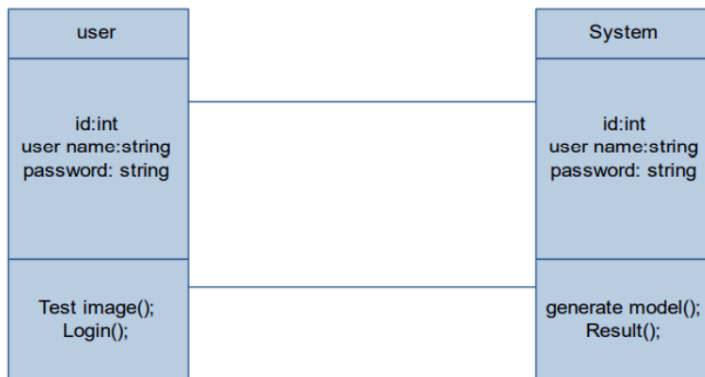
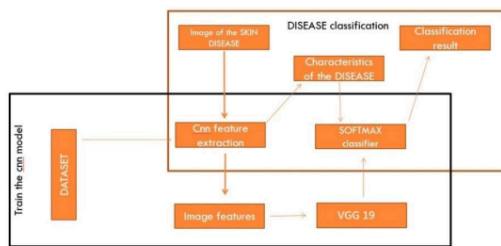


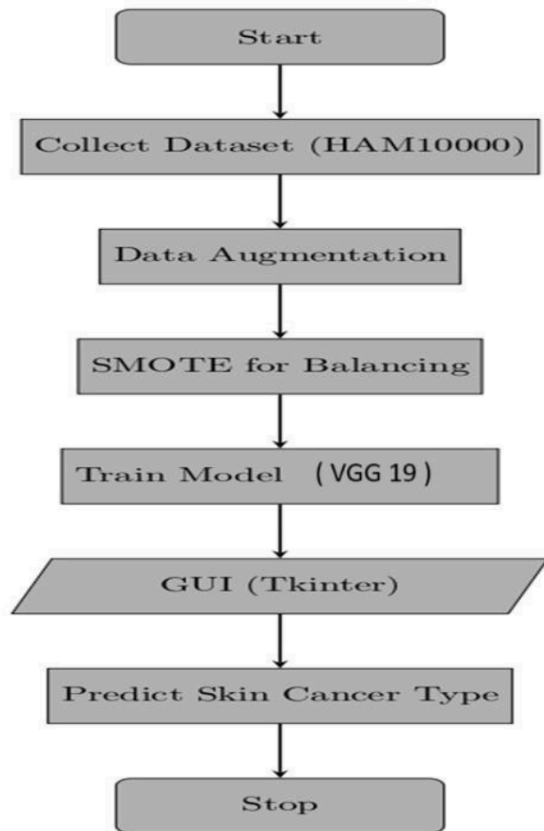
Figure 6: Class Diagram

Fig 3.2: Class Diagram

3.4.2 DATA FLOW DIAGRAM



3.4.3. ACTIVITY DIAGRAM



CHAPTER 4

MODULE DESCRIPTION

11

4.1 DATA COLLECTION AND PREPROCESSING

4.1.1 Dataset and Data Labelling

As a project, the HAM10000 dataset is used. It encompasses more than 9,000 images of dermatoscopic images of ¹²seven types of skin lesions, including melanoma as well as basal cell carcinoma. All images are labeled. This leads to a rather obvious data set that ¹³will be easy to train and test the model on. The model will be effective in use cases outside of its training environment because of the variety of lesion types from several subjects and locations. The skin lesions depicted here are both common and rare, hence important to the proper training of a strong classifier. This dataset is designed for machine learning; therefore, it will be very suitable with deep learning tools. A high-quality image of small details and skin lesions makes feature extraction during training easy.

4.1.2. Data Preprocessing

making sure that the same model is trained on balanced, consistent, and varied data. Preprocessing resampling input data such that it becomes almost ready for efficient and effective training: the images resized to appropriate sizes compatible with the VGG19 architecture; labels organized based on the seven classes of skin lesions. It can mitigate class imbalance by generating synthetic examples of underrepresented lesion types and hence reduce model bias from unbalanced training. Normalization stabilizes training and increases efficiency in learning. Rotation, flipping, and scaling techniques of data augmentation diversify the dataset. These allow the model to manage various inputs encountered in real-world applications. These steps improve the quality of a dataset, thereby

4.2.3 Feature Selection

VGG19, a deep convolutional neural network, extracts hierarchical features from input images by passing them through multiple convolutional layers. These layers detect low-level features like edges, textures, and gradually progress to higher-level features, such as shapes and object parts.

4.2.4 Classification and Model Selection

VGG19, with 19 layers, efficiently handles vanishing gradients and extracts hierarchical features. Pre-trained on ImageNet, it supports transfer learning, reducing training time and boosting accuracy. Its scalability makes it ideal for medical image classification task.

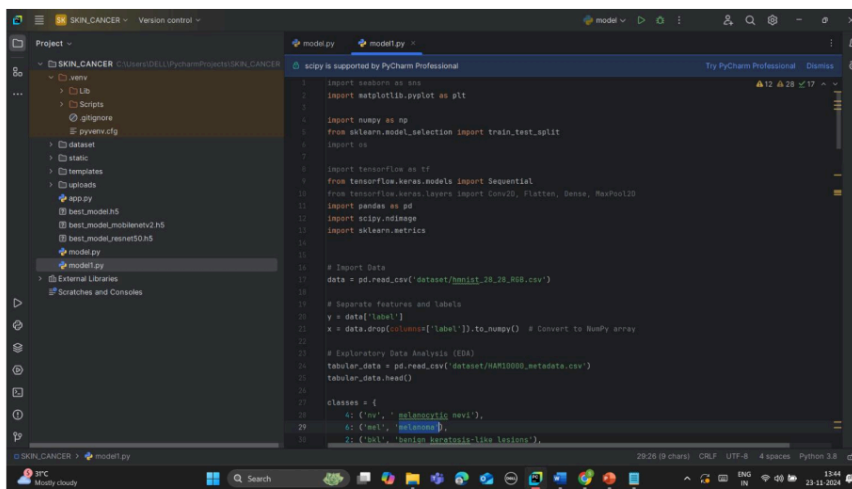
4.2.5. MODEL DEPLOYMENT

An unseen test dataset evaluates ⁶the performance of the model using accuracy, precision, recall, and F1-score metrics. Logarithmic loss computes prediction confidence. Tests on challenging samples such as noisy images, ensuring robustness and reliability.

IMPLEMENTATION AND RESULTS

5.1 IMPLEMENTATION

The implementation begins with the HAM10000 dataset collection, preprocessing (resizing, normalization, SMOTE, and augmentation), and selection of VGG19 for feature extraction and classification. The model undergoes training using transfer learning with optimized settings like batch size and learning rate. It's validated on unseen data using accuracy, precision, recall, and F1-scores. A Tkinter-based GUI enables image upload and instant lesion predictions, providing accessible, real-time skin cancer diagnosis for diverse users.



```
1 import seaborn as sns
2 import matplotlib.pyplot as plt
3
4 import numpy as np
5 from sklearn.model_selection import train_test_split
6 import os
7
8 import tensorflow as tf
9 from tensorflow.keras.models import Sequential
10 from tensorflow.keras.layers import Conv2D, Flatten, Dense, MaxPool2D
11 import pandas as pd
12 import scipy.ndimage
13 import sklearn.metrics
14
15 # Import Data
16 data = pd.read_csv('dataset/ham10000.csv')
17
18 # Separate features and labels
19 x = data['label']
20 y = data['label']
21 x = data.drop(columns=['label']) # Convert to NumPy array
22
23 # Exploratory Data Analysis (EDA)
24 tabular_data = pd.read_csv('dataset/HAM10000_metadata.csv')
25 tabular_data.head()
26
27 classes = {
28     0: ('nev', 'melanocytic nevus'),
29     1: ('bas', 'basaloid'),
30     2: ('hsk', 'benign keratosis-like lesions'),
31 }
```

1 Fig 5.1 Data preprocessing & Feature selection

14

model.py

model1.py

app.py

scipy is supported by PyCharm Professional

Try PyCharm Professional

Dismiss

```

108 # Model Training
109 vgg = tf.keras.applications.VGG19(input_shape=(32, 32, 3), weights='imagenet', include_top=False)
110 for layer in vgg.layers:
111     layer.trainable = False
112
113 model = tf.keras.models.Sequential([
114     vgg,
115     Conv2D(128, (3, 3), padding='same'),
116     tf.keras.layers.Activation('relu'),
117     Conv2D(64, (3, 3), padding='same'),
118     tf.keras.layers.Activation('relu'),
119     Conv2D(32, (3, 3), padding='same'),
120     tf.keras.layers.Activation('relu'),
121     tf.keras.layers.BatchNormalization(),
122     tf.keras.layers.Dropout(0.25),
123     Flatten(),
124     Dense(7, activation='softmax')
125 ])
126
127 print(model.summary())
128
129 callback = tf.keras.callbacks.ModelCheckpoint(
130     filepath='best_model.h5',
131     monitor='val_accuracy', # Corrected from 'val_acc' to 'val_accuracy'
132     mode='max',
133     verbose=1
134 )
135
136 model.compile(loss=sparse_categorical_crossentropy,
137               optimizer='adam',
138               metrics=['accuracy'])

```

Fig 5.2 TRAINING THE DATASET

model.py

model1.py

app.py

scipy is supported by PyCharm Professional

Try PyCharm Professional

Dismiss

```

108 plt.plot(history.history['accuracy'])
109 plt.plot(history.history['val_accuracy'])
110 plt.title('Model Accuracy')
111 plt.xlabel('epoch')
112 plt.legend(['Train', 'Val'], loc='upper left')
113 plt.show()
114
115 plt.plot(history.history['loss'])
116 plt.plot(history.history['val_loss'])
117 plt.title('Model Loss')
118 plt.xlabel('epoch')
119 plt.legend(['Train', 'Val'], loc='upper left')
120 plt.show()
121
122 # Model Testing
123 model.load_weights('best_model.h5')
124
125 loss, acc = model.evaluate(X_test, Y_test, verbose=2)
126
127 y_pred_ohv = model.predict(X_test)
128 y_pred_labels = np.argmax(y_pred_ohv, axis=1)
129
130 confusion_matrix = sklearn.metrics.confusion_matrix(y_true=Y_test, y_pred=y_pred_labels)
131
132 class_labels = ['glass', 'beer', 'bottle', 'apple', 'mug', 'candy', 'milk']
133 report = sklearn.metrics.classification_report(Y_test, y_pred_labels, target_names=class_labels)
134 print(report)

```

Fig 5.3 TESTING

5.2 OUTPUT SCREENSHOTS

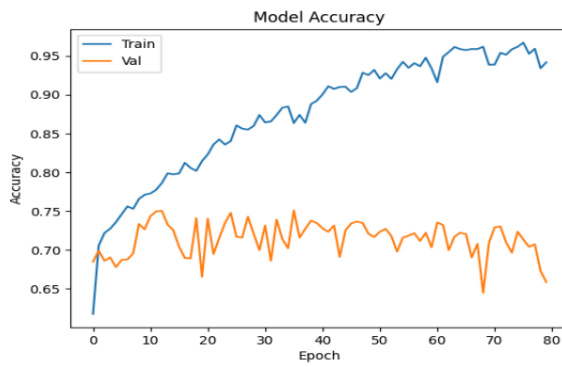


Fig 5.4 MODEL ACCURACY

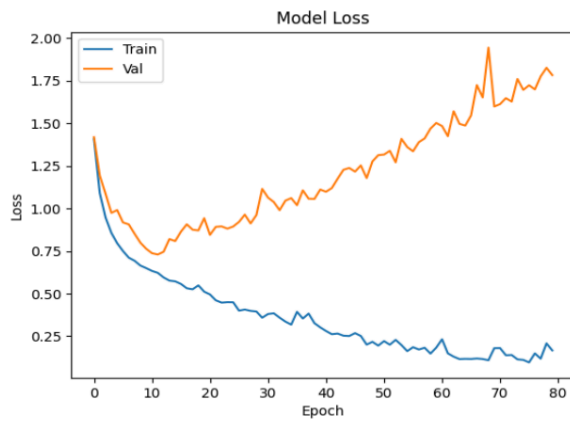


Fig 5.5 MODEL LOSS

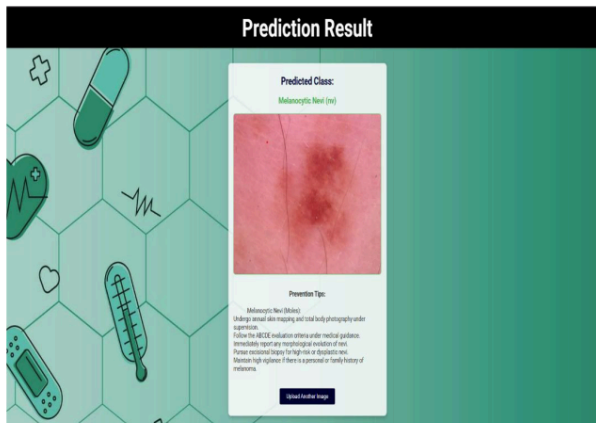


FIG 5.6 PREDICTION OF SKIN CANCER

CONCLUSION AND FUTURE ENHANCEMENT

6.1 CONCLUSION

This research work introduces an advanced skin cancer detection system that uses the VGG19 deep learning model. The proposed system improves on earlier implementations, such as ResNet-50, by using a pre-trained VGG19 network and advanced preprocessing techniques to achieve higher accuracy and reliability in the detection of skin cancer. The use of deep learning allows for automatic classification of skin lesions for efficient and non-invasive early detection. This is especially useful for remote locations where access to dermatological expertise may be limited. The system's ability to analyze images with high precision enhances its potential for widespread use, offering a more accessible method of diagnosing skin cancer. Future developments could include the integration of additional imaging modalities, such as ultrasound and MRI, to further improve diagnostic capabilities. This development may make these assessments even more comprehensive to offer better treatments and management for the disease in skin cancer patients.

6.2. FUTURE ENHANCEMENT

Several key enhancements can be made to further improve the skin cancer detection system's efficiency, usability, and overall impact. First, incorporating real-time monitoring would enable the continuous tracking of skin lesion changes over time. This would allow for early detection of any suspicious developments, ensuring timely intervention and reducing the chances of misdiagnosis. Developing a mobile application is another significant improvement that could broaden accessibility. The system is user-friendly; people can capture images of their skin lesions easily using their smartphone cameras, and it makes the system more convenient for users, especially in remote or underserved areas where access to dermatological care is limited. Increasing the variety of dermoscopic images in the dataset would be essential to further increase the robustness of the model. Images would be drawn

from different skin types, conditions, and lesion stages, and the model would be more practical in finding a range of cutaneous carcinomas across different groups of people. Additionally, implementing genetic-based analysis could provide a more personalized skin cancer risk assessment. By factoring in genetic predispositions, the system could offer tailored recommendations and early detection for individuals who may be at higher risk due to hereditary factors. Finally, integrating Explainable AI (XAI) techniques would help improve transparency and build trust in the model's predictions. By providing clear and understandable reasons behind the system's decisions, users and healthcare professionals can feel more confident in the results.

These developments would allow AI-based skin cancer diagnosis to be increasingly effective, providing reliable, accessible, and personalized applications ² in the field of modern health care, ultimately leading to improved patient outcomes and a tremendous decrease in skin cancer morbidity and mortality.

REFERENCES

- [1] S. Gupta, "Biases in Machine Learning Affecting Generalization," *in Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2018.
- [2] K. Lee, "Types of Bias in Machine Learning Models," *in Proc. IEEE Int. Conf. on Machine Learning and Applications (ICMLA)*, 2021.
- [3] A. Patel, "Anchoring Bias and Its Impact on Model Adaptability," *in Proc. IEEE Int. Conf. on Data Mining (ICDM)*, 2022.
- [4] M. Johnson, "Selection Bias and Its Effects on Model Fairness," *in Proc. IEEE Int. Conf. on Artificial Intelligence and Statistics (AISTATS)*, 2023..
- [5] L. Chen, "Measuring Inefficiencies and Biases in Convolutional Neural Networks," *in Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2022.
- [6] T. Nguyen, "Evaluating Biases Across CNN Architectures and Metrics for Measurement," *in Proc. IEEE Int. Conf. on Machine Learning (ICML)*, 2023.
- [7] R. Patel, "Improving Bias Understanding and Reduction Strategies in CNNs for Enhanced Model Performance and Fairness," *in Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2024.
- [8] P. Kumar and P. Yashini, "A Novel Machine Learning Approach for Medical Recommendation System," 2024 10th International Conference on Communication and Signal Processing (ICCSP), Melmaruvathur, India, 2024, pp. 1-6, doi: 10.1109/ICCSP60870.2024.10544269.
- [9] P. Kumar, V. Subathra, Y. Swasthika and V. Vishal, "Disease Diagnosis Using

Machine Learning on Electronic Health Records," 2024 International Conference on Communication, Computing and Internet of Things (IC3IoT), Chennai, India, 2024, pp. 1-6, doi: 10.1109/IC3IoT60841.2024.10550235.

[10] S. Thompson, et al., "Limitations of Elementary Metrics in Evaluating Model Performance and Bias," *in Proc. IEEE Int. Conf. on Pattern Recognition (ICPR)*, 2021.

[11] A. Rodriguez, et al., "Ineffective Training Procedures Leading to Biased Models with Poor Generalization," *in Proc. IEEE Int. Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2023.

[12] M. Williams, et al., "Impact of Lack of Resilience on Model Dependability in Real-World Settings," *in Proc. IEEE Int. Conf. on Artificial Intelligence (IJCAI)*, 2022.

[13] J. Liu, et al., "The Consequences of Uncorrected Bias on Model Accuracy and Fairness," *in Proc. IEEE Int. Conf. on Machine Learning (ICML)*, 2021.

[14] K. Singh, et al., "Proactive Bias Detection and Correction for Reliable and Fair Models," *in Proc. IEEE Int. Conf. on Computer Vision (ICCV)*, 2023.

[15] L. Zhang, et al., "Post-Training Analysis in CNNs for Bias Detection and Quantification," *in Proc. IEEE Int. Conf. on Computer Vision and Pattern Recognition (CVPR)*, 2022.

[16] A.D. Selbst, "Fairness and Abstraction in Sociotechnical Systems," in Proc. FAT Conference*, 2019.

[17] B.H. Zhang, "Mitigating Unwanted Biases with Adversarial Learning," in Proc.

Neural Information Processing Systems (NeurIPS), 2018.

[18] N. Kallus, and S. Athey, "Real-time Bias Mitigation in Machine Learning Models," in Proc. Int. Conf. on Machine Learning (ICML), 2018.

[19] P. Hoffman, "Tackling Dataset Bias With an Automated Collection of Real-World Samples," *in Proc. IEEE Int. Conf. on Data Science and Advanced Analytics (DSAA)*, 2021.

[20] L. Wright, "The Opacity of Artificial Intelligence Makes It Hard to Tell When Decision-Making Is Biased," *in Proc. IEEE Int. Conf. on Artificial Intelligence (ICAI)*, 2020.

ORIGINALITY REPORT

20%

SIMILARITY INDEX

13%

INTERNET SOURCES

3%

PUBLICATIONS

15%

STUDENT PAPERS

PRIMARY SOURCES

1 Submitted to University of Illinois at Urbana-Champaign 14%
Student Paper

2 Danish Jamil, Farheen Qazi, Dur-E-Shawar Agha, Sellappan Palaniappan. "Classification of skin lesion using deep convolutional neural network by applying transfer learning", Journal of Autonomous Intelligence, 2023 1%
Publication

3 Submitted to University College London 1%
Student Paper

4 ebin.pub 1%
Internet Source

5 Submitted to La Trobe University <1%
Student Paper

6 www.ijraset.com <1%
Internet Source

7 www.supsalv.org <1%
Internet Source

8 baadalsg.inflibnet.ac.in <1%
Internet Source

9 users.ecel.ufl.edu <1%
Internet Source

10 ethesis.nitrkl.ac.in <1%
Internet Source

11 www.mdpi.com <1%
Internet Source

12 www.newswire.ca

Internet Source

<1 %

13 [dokumen.pub](#)

Internet Source

<1 %

14 [www.dgct.ac.in](#)

Internet Source

<1 %

15 [www.ijitee.org](#)

Internet Source

<1 %

16 Samarjeet Borah, Ratna Raja Kumar Jambi, Sharifah Sakinah Syed Ahmad, Mahendra Prabhakar Deore. "Applied Soft Computing Techniques - Theoretical Principles and Practical Applications", Apple Academic Press, 2025
Publication

<1 %

Exclude quotes On

Exclude matches Off

Exclude bibliography On